Problem Set 6

Due date: Electronic submission of the pdf file of this homework is due on **3/7/2025 before 11:59pm** on canvas.

Name: (put your name here)

Resources. (All people, books, articles, web pages, etc. that have been consulted when producing your answers to this homework)

On my honor, as an Aggie, I have neither given nor received any unauthorized aid on any portion of the academic work included in this assignment. This work is my own. Furthermore, I have disclosed all resources (people, books, web sites, etc.) that have been used to prepare this homework. Looking up solutions is prohibited.

Signature: _____

Make sure that you describe all solutions in your own words, and that the work presented is your own! Typesetting in LaTeX is required. Read the chapter on amortized analysis in our textbook.

Problem 1 (20 points)**.** Suppose that you have a row of coins with values that are positive integers $v_1, v_2, \ldots, v_n$. These values are not necessarily distinct. The goal is to pick up the highest total value of coins as possible. However, one cannot pick up two coins that are adjacent to each other. (a) Describe an algorithmic solution that is based on the dynamic programming principle. Carefully explain why your solution is correct. (b) What is the solution for coins with values (25,5,5,25,10,10,5,25,1).

Solution.

Problem 2 (10 points)**.** In the lecture, we discussed a stack with PUSH, POP, and MULTIPOP operations. If the set of stack operations included a MULTI-PUSH operation, which pushes k items onto the stack, would the $O(1)$ bound on the amortized cost of stack operations continue to hold?

Solution.

Problem 3 (10 points)**.** In the lecture, we discussed a k-bit counter with an INCREMENT operation. Show that if a DECREMENT operation were included in the k-bit counter example, n operations could cost as much as $\Theta(nk)$ time.

Solution.

Problem 4 (20 points)**.** Suppose we perform a sequence of n operations on a data structure in which the i-th operation costs i if i is an exact power of 2, and 1 otherwise. Use aggregate analysis to determine the amortized cost per operation.

Solution.

Problem 5 (20 points)**.** Redo Problem 4 using an accounting method of analysis.

Solution.

Problem 6 (20 points)**.** Redo Problem 4 using a potential method of analysis.

Solution.

Discussions on canvas are always encouraged, especially to clarify concepts that were introduced in the lecture. However, discussions of homework problems on canvas should not contain spoilers. It is okay to ask for clarifications concerning homework questions if needed. Make sure that you write the solutions in your own words.

Checklist:

- ☐ Did you add your name?
- ☐ Did you disclose all resources that you have used?
  (This includes all people, books, websites, etc. that you have consulted)
- ☐ Did you sign that you followed the Aggie honor code?
- ☐ Did you solve all problems?
- ☐ Did you submit the pdf file of your homework?